Python Assignment
Topic-3: Core Python Concepts

Q1. Understanding data types: integers, floats, strings, lists, tuples, dictionaries, sets.

Ans.

Data types define the type of value stored in a variable. Python supports many data types which make programming flexible. Integer is used for whole numbers like 10 or 20. Float is used for decimal numbers like 10.5. String is used to store text like names. List is used to store multiple values in one variable. Tuple is similar to list but it cannot be changed. Dictionary stores data in key and value pairs. Set is used to store unique values. These data types help in organizing and managing data easily in Python programs.

Example:

```
a = 10
b = 10.5
name = "Shivani"
list1 = [1,2,3]
tuple1 = (1,2,3)
dict1 = {"name":"Shivani"}
```

Q2. Python variables and memory allocation.

Ans.

Variables are used to store data in Python. When we assign a value to a variable, Python automatically allocates memory for it. We do not need to declare the type of variable in Python. The type is decided automatically based on the value. Variables act as containers that hold data values. Python uses dynamic typing which means the same variable can store different types of values at different times. Memory allocation is handled automatically by Python, so programmers do not need to manage memory manually.

Example:

```
x = 10
x = "Hello"
print(x)
```

Q3. Python operators: arithmetic, comparison, logical, bitwise.

Ans.

Operators are used to perform operations on variables and values. Arithmetic operators are used for basic calculations like addition, subtraction, multiplication, and division. Comparison operators are used to compare values and return true or false. Logical operators are used to combine conditions like and, or, and not. Bitwise operators are used to perform operations on binary numbers. These operators help in performing different types of tasks in Python programs. They are very important for writing logic and conditions in coding.

Example:

```
a = 10
b = 5
print(a + b)
print(a > b)
print(a > 5 and b < 10)
```